

Part 4

BAYESIAN MODEL CHECKING

Spring 2026

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4.1 – Model comparison versus model checking

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To identify an appropriate model for a given dataset, we can employ two strategies.

- Formal model comparison: list all candidate models, assign a prior probability to each, and compute their posterior probabilities, to select the winner or use a weighted average of all candidates.
- Posterior predictive checks: evaluate models individually by determining if it is plausible that a model could generate data similar to what was observed in practice.

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The model comparison approach is highly structured and fits neatly within the standard inferential framework.

However, it only provides a relative ranking, and it does not tell you if the winning model is actually a good fit for data.

It offers no diagnostics regarding model failure nor any guidance on how to improve the existing candidates.

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Predictive checks provide the necessary hints to uncover what is wrong with a model, allowing for an iterative process of model criticism and improvement.

This ensures that the final model is not just best among a set of alternatives, but is good enough for the task at hand.

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4.2 – Model checking through predictive checks

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Bayesian models can work as data simulators through their predictive distributions.

Building a Bayesian model is like building a data simulator.

The most natural way to assess a model is to evaluate the quality of the synthetic data it generates.

By checking how well a model replicates the observed reality, we can identify its specific shortcomings.

That helps discover how to improve it.

The goal is to compare samples simulated from predictive distributions, y^{rep} , with the real observed data, y .

We ask ourselves whether the simulated data and the real data are indistinguishable or not.

Complex data sets can be assessed according to many different criteria, and one needs to focus the comparison on those features of reality that one cares the most about.

Real-world systems are complex.

A model might succeed in some areas but fail in others.

We must focus our comparison on the specific features that are most relevant to our scientific goals.

In practice, that means comparing data sets based on location, spread, skewness, frequency of extreme values, or other more problem specific features.

While less formally structured, predictive checks follow a three-step process:

1. Choose specific statistics $T(y)$ or graphical displays that best summarize the data features of interest.
2. Select the appropriate predictive distribution to generate multiple sets of replicated data.
3. Determine the degree of conflict between the observed $T(y)$ and the distribution of $T(y^{\text{rep}})$.

Example 8:

Puig, X., and Ginebra, J., (2014). "A Bayesian cluster analysis of election results," *Journal of Applied Statistics*, 41, 73-94.

<https://doi.org/10.1080/02664763.2013.830088>

A Bayesian cluster analysis of election results

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A Bayesian cluster analysis for the results of an election based on multinomial mixture models is proposed. The number of clusters is chosen based on the careful comparison of the results with predictive simulations from the models, and by checking whether models capture most of the spatial dependence in the results. By implementing the analysis on five recent elections in Barcelona, the reader is walked through the choice of the best statistics and graphical displays to help choose a model and present the results. Even though the models do not use any information about the location of the areas in which the results are broken into, in the example they uncover a four-cluster structure with a strong spatial dependence, that is very stable over time and relates to the demographic composition.

Keywords: Bayesian hierarchical model; model checking; predictive checks; spatial data

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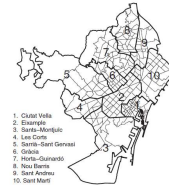


Figure 1. Map of Barcelona, divided into 10 districts and 248 smaller areas.

Table 1. Part of the 248 × 8 table with the 2003 results.

District	Area	CIU	PSC	PPC	ERC	ICV	Others	b + n	abs	N
1	1	195	375	76	86	58	4	15	701	1510
1	2	208	333	75	97	70	20	6	790	1599
1	3	226	307	70	133	54	8	11	635	1444
10	247	471	1489	555	235	195	50	48	2185	5228
10	248	441	1535	592	245	229	48	34	2202	5326
Total		227,783	249,020	123,163	126,626	69,234	9813	9482	407,294	1,222,415

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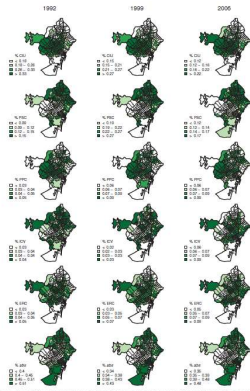


Figure 2. Maps with the percentage of vote for the main parties and of the abstention in the 1992, 1999 and 2006 elections. Percentages are categorized according to their quartiles to help emphasize the spatial dependence in the results.

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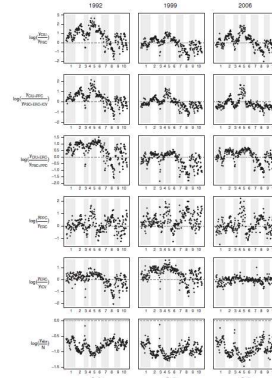


Figure 3. Value taken by DP (y), DP (x), DP (y), DP (x) and DP (y) in the 1992, 1999 and 2006 elections at the 248 areas considered, grouped by district.

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Box 1. Bayesian multinomial s-cluster model.

$$\begin{aligned} (y_1, \dots, y_{248}) \mid (\theta_1, \dots, \theta_s), \zeta &\sim \prod_{i=1}^{248} \text{Mult}(N_i, \theta_{\zeta_i}) \\ (\theta_1, \dots, \theta_s) &\sim \prod_{r=1}^s \text{Dirichlet}(a_{r1}, \dots, a_{rk}) \\ \pi(\zeta_1, \dots, \zeta_{248} \mid \omega) &= \prod_{i=1}^{248} \omega_{\zeta_i} \\ (\omega_1, \dots, \omega_s) &\sim \text{Dirichlet}(b_1, \dots, b_s) \end{aligned}$$

Box 2. Bayesian Dirichlet-multinomial s-cluster model.

$$\begin{aligned} (y_1, \dots, y_{248}) \mid (\mu_1, \dots, \mu_s), (\tau_1, \dots, \tau_s), \zeta &\sim \prod_{i=1}^{248} \text{Dir-Mult}(N_i, \tau_{\zeta_i}, \mu_{\zeta_i}) \\ (\mu_1, \dots, \mu_s) &\sim \prod_{r=1}^s \text{Dirichlet}(m_{r1}, \dots, m_{rk}) \\ (\tau_1, \dots, \tau_s) &\sim \prod_{r=1}^s \text{Gamma}(c_r, d_r) \\ \pi(\zeta_1, \dots, \zeta_{248} \mid \omega) &= \prod_{i=1}^{248} \omega_{\zeta_i} \\ (\omega_1, \dots, \omega_s) &\sim \text{Dirichlet}(b_1, \dots, b_s) \end{aligned}$$

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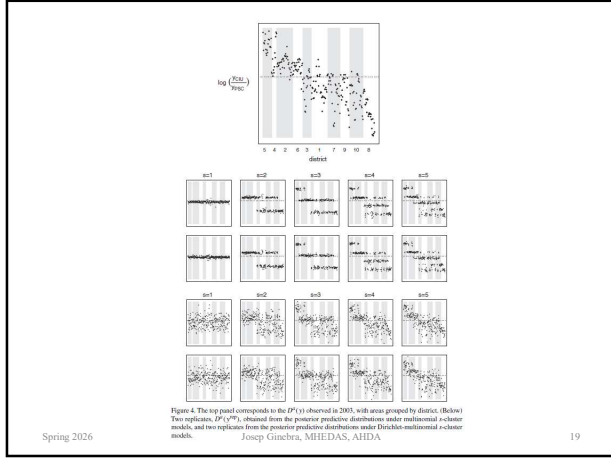
- Multinomial 1-cluster model
- Multinomial 2-cluster model
- Multinomial 3-cluster model
- Multinomial 4-cluster model
- Multinomial 5-cluster model

- Dirichlet-multinomial 1-cluster model
- Dirichlet-multinomial 2-cluster model
- Dirichlet-multinomial 3-cluster model
- Dirichlet-multinomial 4-cluster model
- Dirichlet-multinomial 5-cluster model

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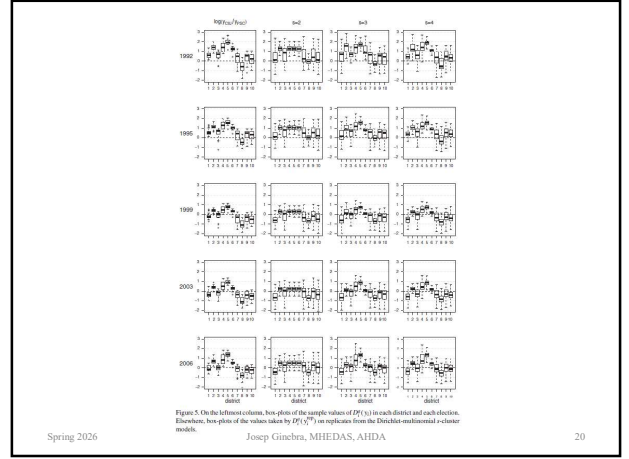
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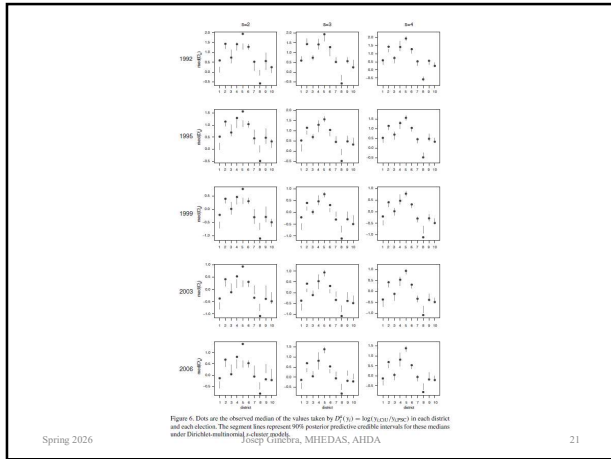
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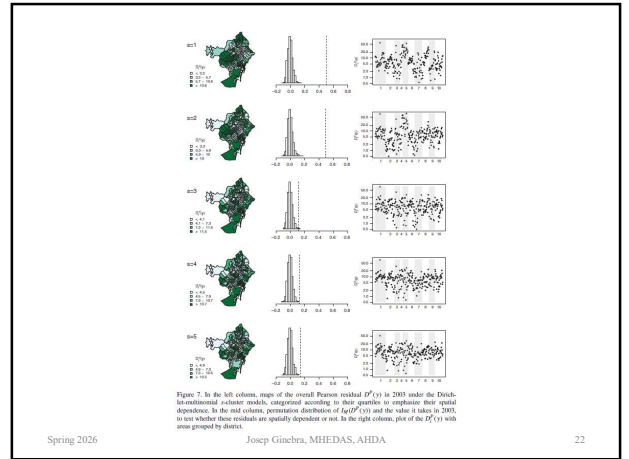
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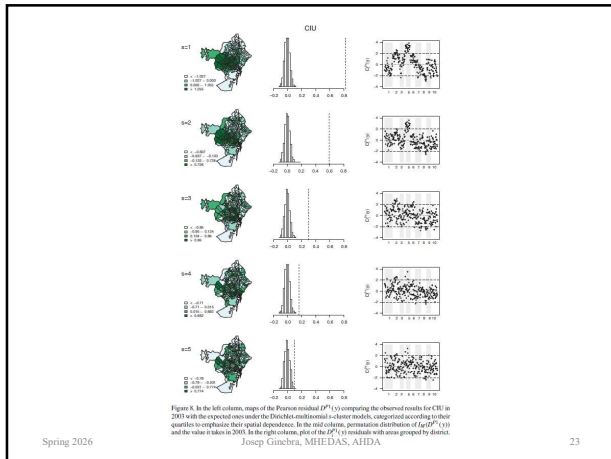
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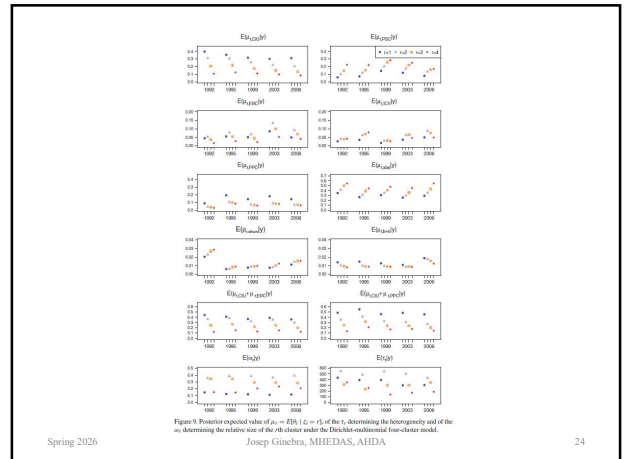
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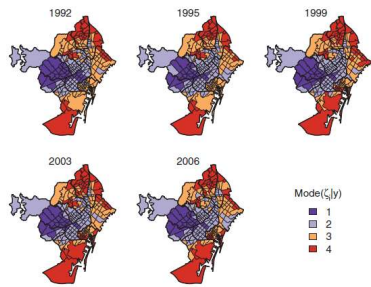


Figure 10. Classification of the 248 areas into four clusters under the Dirichlet-multinomial four-cluster model.